Module 2 Assignment 2

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Assessing Accuracy

Assignment Details

Purpose

The goal of this assignment is to use R to calculate and understand accuracy of estimates through standard error and confidence intervals.

Task

Write R code to successfully answer each question below or write text to successfully answer the question.

Criteria for Success

- Code is within the provided code chunks
- Code chunks run without errors
- Code produces the correct result
- Text answers correctly address the question asked

Due Date

March 23 before lab

Assignment Questions

Each question is worth 1 point unless otherwise noted.

Set-up

For our assignment today, we will be using data from my PhD field site in southeastern Arizona in the Chiricahuas. In fact, we will be using some of the first data I ever collected for the project! To learn more about the Portal Project, check out the project's website. Researchers have been collecting data at this site monthly since 1977!

What you need to know for today is that this is small mammal (rodent) data.

The site has 24 plots, although we will only be using 18 of them today (6 of them are rodent removal plots, so they aren't very helpful in estimating rodent densities!).

As such, `n` will be 18 for this assignment. We do not have a `N`.

Download the Data

The first thing we need to do is download the data. My PhD lab group actually created our own R package for using the cleaned up data, so we are going to load that package, called `portalr`.

```
# if a message pops up when you run this code chunk, you can ignore it!
library(portalr)
```

Now that we have loaded the packages we need, we are going to download some rodent abundance data. This `abundance()` function downloads the entire dataset, so it might take a few minutes.

Let's take a look at what the dataset looks like.

```
head(abund)
```

```
##   period treatment plot BA DM DO DS NA OL OT PB PE PF PH PI PL PM PP RF RM RO
## 1     27   control    1  0  0  0  0  0  0  1  0  0  0  0  0  0  0  1  0  0  0
## 2     27   control    2  0  2  0  0  0  0  2  0  0  1  0  0  0  0  0  0  0  0
## 3     27   control    4  0  1  0  2  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## 4     27   control    6  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## 5     27   control    8  0  0  0  1  0  0  0  0  0  0  0  0  0  0  1  0  0  0
## 6     27   control    9  0  1  0  2  0  1  0  0  0  0  0  0  0  0  0  0  0  0
##   SF SH SO
## 1  0  0  0
## 2  0  0  0
## 3  0  0  0
## 4  0  0  0
## 5  0  0  0
## 6  0  0  0
```

The data has columns with information about the trapping session and plots; the remaining columns with 2 letters each represent rodent species. The values in the columns indicate how many individuals of that species were caught in a given plot.

Subsetting the Data

We want to choose data only from a certain data collection period and only for certain species. To do this, we are going to filter the data to select trapping period "440" (the 440th time the site was trapped–this was back in 2015. We are over 500 trapping sessions now!) and get rid of the rodent removal plots.

We are also going to select only the columns that are relevant to us.

For this assignment, we are going to be calculating values for two different species of pocket mice: *Chaetodipus penicillatus* and *Chaetodipus baileyi*. In the Portal data, these species are abbreviated as `PP` and `PB`, respectively.

When we choose which columns we keep, we will keep the period, plot, and the PP and PB columns. These columns have the number of each species caught on each plot that trapping session.

```
# filter period 440, exclude "removal" treatment type, and select "PP" and "PB" column
abund440 <- abund[abund$period == 440 & abund$treatment != "removal",
                  c("period", "plot", "PP", "PB")]
```

The `abund440` object you just created by running the code chunk above is the data frame that we will use for the rest of the assignment.

Estimating the Density of *Chaetodipus pencicillatus* (PP)

The first bit of this assignment is very similar to Module 2 Assignment 1, so if you're feeling stuck with the code, you might want to check that assignment out.

As a reminder, when we are estimating a *density*, we are estimating the *mean* number of entities per sample unit.

1. First things first, we want to create an object for our sample size. We don't know `N` in this scenario, but we do know `n`. Create the object `n`, which is the number of sample units that make up our sample. This will be the same for both species.

   To do this, use the `nrow()` function, which counts up the number of rows in a dataframe. The argument for `nrow()` is the name of the dataframe.

```
n <- nrow(abund440)
n
```

```
## [1] 18
```

Calculating Estimates of Population Parameters

We are going to start by calculating some population parameter estimates and measures of uncertainty for the PPs.

2. Create an object called `PP_mean` that contains the sample mean for the PPs

```
PP_mean <- mean(abund440$PP)
PP_mean
```

```
## [1] 11.05556
```

3. Create an object called `PP_var` which contains the sample variance. Hint: use the `var()` function.

```
PP_var <- var(abund440$PP)
PP_var
```

```
## [1] 15.70261
```

Calculating Standard Error for the Mean

Reminder: because we don't know what `N` is, we don't need to use the population correction factor when calculating our standard error and whatnot.

4. Calculate the variance of the estimate of the mean for the PPs.

```
PP_var_ybar <- PP_var / n
PP_var_ybar
```

```
## [1] 0.8723675
```

5. Calculate the SE of the estimate of the mean.

```
PP_SE <- sqrt(PP_var_ybar)
PP_SE
```

```
## [1] 0.9340061
```

Calculating Confidence Intervals for the Mean

Hooray, we've now calculated the sample mean, sample variance, and the standard error! Let's calculate a 95% confidence interval for our sample mean.

6. Create an object called `alpha95` to represent our alpha value for a 95% confidence interval. Because this is a constant, we can assign a number to it (take a look back at lecture notes).

```
alpha95 <- 0.05
alpha95
```

```
## [1] 0.05
```

7. Create an object called `df` for degrees of freedom. Degrees of freedom in this case are equal to the number of sample units in our sample (`n`) minus 1. Use the object `n` in your calculation.

```
df <- n-1
df
```

```
## [1] 17
```

We are going to use a function called `qt()` to find the t-value that we need. Assuming you've created the objects `alpha95` and `df` above, running this code chunk will give us our t-value, or critical value, for 95% CI.

```
t95 <- qt(1-(alpha95)/2, df)
t95
```

```
## [1] 2.109816
```

8. Using `t95` and the standard error that we calculated earlier, create an object called **`half_width`** which contains the half-width value for the 95% CI.

```
halfwidth_95 <- t95 * PP_SE
halfwidth_95
```

```
## [1] 1.970581
```

9. Calculate the upper and lower limits of the 95% confidence intervals. Use objects you've already created to calculate it.

```
upperCI_95 <- PP_mean + halfwidth_95
lowerCI_95 <- PP_mean - halfwidth_95

upperCI_95
```

```
## [1] 13.02614
```

```
lowerCI_95
```

```
## [1] 9.084975
```

10. What happens when we create 90% CI for the data? Let's calculate 90% CI. (3 points total)

a. Input the correct value for `alpha90` and then run the code chunk to get the new t-value, `t90`.

```
alpha90 <- 0.10
alpha90
```

```
## [1] 0.1
```

```
t90 <- qt(1-(alpha90)/2, df)
t90
```

```
## [1] 1.739607
```

b. Now, using `t90`, calculate the 90% CI.

```
halfwidth_90 <- t90 * PP_SE

upperCI_90 <- PP_mean + halfwidth_90
lowerCI_90 <- PP_mean - halfwidth_90

upperCI_90
```

```
## [1] 12.68036
```

```
lowerCI_90
```

```
## [1] 9.430752
```

c. Compare the 90% CI to the 95% CI. Which one is wider? Why would expect that to be the case?

Answer: 95% CI are wider; they are wider because we are willing to accept less uncertainty, so the range of values that must be included needs to be broader to ensure that we have a higher percentage (95 vs 90) of having the true parameter value included in the range.

Estimating the Density of *Chaetodipus baileyi* (PB)

Let's do the same thing as above for our *Chaetodipus baileyi* (PB) data.

Calculating Estimates of Population Parameters

11. Create an object called `PB_mean` that contains the sample mean.

```
PB_mean <- mean(abund440$PB)
PB_mean
```

```
## [1] 0.6666667
```

12. Create an object called `PB_var` which contains the sample variance. Hint: use the `var()` function.

```
PB_var <- var(abund440$PB)
PB_var
```

```
## [1] 1.647059
```

Calculating Standard Error for the Mean

13. Start by calculating the variance of the estimate of the mean

```
PB_var_ybar <- PB_var / n
PB_var_ybar
```

```
## [1] 0.09150327
```

14. Calculate the SE of the estimate of the mean.

```
PB_SE <- sqrt(PB_var_ybar)
PB_SE
```

```
## [1] 0.3024951
```

Calculating Confidence Intervals

15. Create 95% CI for the PB data. You should already have the t-value in your environment, so you don't need to go through calculating that again.

```
halfwidth_95_PB <- t95 * PB_SE

upperCI_95_PB <- PB_mean + halfwidth_95_PB
lowerCI_95_PB <- PB_mean - halfwidth_95_PB

upperCI_95_PB
```

```
## [1] 1.304875
```

```
lowerCI_95_PB
```

```
## [1] 0.02845785
```

Interpretation

16. Compare the estimated densities of PPs and PBs. Which species is present in higher densities at the site?

Answer: PPs are at a higher density (`PP_mean` is higher than `PB_mean`)

17. Now, compare the ranges of the 95% CI for the PP and the PB data. Which density estimate seems to be more accurate? How are you using to make that determination?

    Taking a look back at the equation for calculating confidence intervals and the values for each species in that equation, what value is driving the difference in how wide the confidence intervals (AKA the range of the interval)?

Answer: PB seems more accurate because the range of the confidence intervals is narrower than those for PPs. This is because the SE is lower, leading to smaller CIs. [Note: this is actually being driven by the differences in standard deviation since the sample sizes are the same, but they don't need to say that to get credit]